

INCITE Report and Renewal Contribution

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1 Project Achievements

1.1 Significant Accomplishments to Date

This INCITE allocation has produced a working first principles framework for self trapped excitons (STEs), along with the main calculations needed to apply and test it. The central advance is a practical link between excited-state forces and exciton self-energy theory. This link makes the STE relaxation problem more stable and gives access to dynamical quantities that are not available from forces alone.

The main accomplishments are:

- A self consistent exciton self-energy implementation was developed for STE analysis, including tadpole and Fan Migdal contributions.
- Finite momentum exciton data were generated for LiF, SiO₂, and Cs₂AgInCl₆ across the main grid sizes needed for convergence and extrapolation.
- The self-energy distortion provided an effective starting point for the excited-state forces method in Cs₂AgInCl₆, reducing the number of costly trial force iterations.
- The work established a direct comparison between harmonic self-energy theory and nonlinear relaxation with excited-state forces.
- The method now gives access to phonon resolved contributions, temperature dependent emission broadening, and first principles estimates of the Stokes shift.

The broader significance is that the project has moved from method setup to a usable route for temperature dependent STE photoluminescence from first principles. This is difficult because STEs combine strong exciton phonon coupling, lattice relaxation, large distortions, and optical spectra. Excited-state forces can describe the relaxed structure, but they do not directly give dynamical temperature dependence. The self-energy method fills that gap and improves the starting point for the force calculation.

The calculation campaign is already broad. Benchmark calculations were completed on LiF and SiO₂, which have prior STE studies and, for SiO₂, useful experimental reference data. The same workflow was

then extended to the target double perovskite, $\text{Cs}_2\text{AgInCl}_6$. For each material, finite momentum exciton calculations were performed on $2 \times 2 \times 2$, $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grids, followed by extrapolation toward the dense grid limit. The inputs came from DFT, DFPT, EPW, and BerkeleyGW *GW* plus Bethe Salpeter equation calculations.

The combined method was motivated by the $\text{Cs}_2\text{AgInCl}_6$ production runs. Repeated iterations in the excited-state forces method were costly, and some trial distortions led to slow or oscillatory convergence. This suggests that the STE energy landscape can contain barriers or competing local minima. Toy models with Fröhlich type coupling support the same picture: STE formation balances the momentum dependent kinetic energy of the exciton against the energy gained from lattice distortion. A good initial distortion is therefore part of the physics of the calculation, not only a numerical convenience.

To address this, we developed an exciton self-energy approach based on tadpole and Fan Migdal diagrams. The tadpole term was solved self consistently. The resulting lattice distortion then served as a physically motivated starting point for the excited-state forces method. The two methods are complementary. The self-energy method captures harmonic lattice response and dynamical effects. The excited-state forces method captures the nonlinear static relaxation. Together, they reduce the dependence on arbitrary starting distortions.

The remaining work is now mainly analysis. The completed calculations motivate two follow on questions. What barriers control STE formation in $\text{Cs}_2\text{AgInCl}_6$? When is harmonic self-energy theory sufficient, and when is full nonlinear relaxation needed? Comparing LiF, SiO_2 , and $\text{Cs}_2\text{AgInCl}_6$ will help answer both questions.

1.2 Allocation Use

About 1,300 node hours have been used to date by `krishnaa423`. The time supported the main calculation campaign: finite momentum exciton inputs, development and tests of the STE self-energy code, production self-energy runs, and calculations with excited-state forces for $\text{Cs}_2\text{AgInCl}_6$. Usage was not uniform because the work alternated between production runs, code validation, and analysis.

The approximate breakdown is:

- About 45 percent for DFT, DFPT, EPW, *GW*, and BSE input generation for LiF, SiO_2 , and $\text{Cs}_2\text{AgInCl}_6$.
- About 25 percent for code development, debugging, and benchmarking, mainly using LiF as the first test case.
- About 20 percent for production self-energy calculations and convergence studies.
- About 10 percent for calculations with excited-state forces in $\text{Cs}_2\text{AgInCl}_6$, including tests of the self-energy based initial distortion.

Job sizes varied by calculation type. BerkeleyGW finite momentum exciton calculations were batched over momentum points and grid sizes. The self-energy calculations used distributed linear algebra and parallelism over momentum points. Memory needs were set by matrices with dimensions of roughly 10^5 for normal cell calculations and up to roughly 10^7 effective degrees of freedom for larger supercell related objects. Frontier’s 512 GB per node was important for distributing these data.

For the rest of the current allocation year, additional time will be used mainly for final checks, analysis driven reruns, and any targeted $\text{Cs}_2\text{AgInCl}_6$ refinement needed after inspecting the current results. Related materials or defect and doping configurations may be tested if the analysis identifies a clear need. Usage will remain intermittent, since large batches will be launched only after the grid, structure, and initial distortion are checked.

1.3 Application Parallel Performance

The production work used established electronic structure codes together with the project specific self-energy code. Quantum ESPRESSO and EPW were used for DFT, DFPT, and electron phonon inputs. BerkeleyGW was used for *GW* and finite momentum BSE calculations. BerkeleyGW fits the Frontier workflow well because

the finite momentum calculations can be split into independent batches, and the expensive *GW* and BSE stages expose parallelism over bands, transitions, and momentum points.

The main progress for the self-energy code was a distributed implementation for Frontier. PETSc, SLEPc, petsc4py, and slepc4py were compiled with ROCm and HIP support. This allowed distributed arrays and matrices to be built on Frontier and allowed SLEPc eigensolvers to be used for diagonalization. The self-energy calculation was also parallelized over q points, which made the $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grid calculations more tractable.

The largest performance gain was algorithmic. The self-energy method reduces wasted iterations in the excited-state forces method by providing a better initial distortion. The implementation also distributes the expensive matrix and momentum dependent operations across the machine. This is important for $\text{Cs}_2\text{AgInCl}_6$, where each iteration with excited-state forces can require DFT, DFPT, *GW*, and BSE calculations.

The main remaining challenge is DFPT. We have not yet identified a Quantum ESPRESSO build that uses Frontier’s AMD GPUs effectively for this part of the calculation. As a result, input generation remains less accelerated than the BerkeleyGW and self-energy stages. This was anticipated, since GPU support is not uniform across the full electronic structure stack. Addressing it would require a large software effort outside the main scope of this STE theory work.

1.4 Data Storage

The current data volume is estimated to be a few terabytes. Most of this comes from wave functions, dielectric matrices, electron phonon data, *GW* and BSE intermediate files, and self-energy outputs for dense grids. The largest storage pressure comes from $\text{Cs}_2\text{AgInCl}_6$ and from supercell related excited-state forces data.

By the end of the current project period, the stored data should remain in the several terabyte range if only essential production data are retained. Scratch files that can be regenerated will be deleted. The retained data will include inputs, final outputs, convergence data, and selected intermediate files needed for reproducibility. Completed calculation directories will be compressed and moved to long term storage using center supported tools such as `hsi`. Publication data will be reduced to portable files containing structures, phonon mode projections, exciton energies, self-energy contributions, spectra, and figure scripts.

2 Project Plans for Next Year

2.1 Summary of Project Plans

The next allocation period will focus on turning the completed calculation campaign into final scientific results. The main goal is to connect the existing first principles calculations to STE observables: the relaxed distortion, Stokes shift, emission linewidth, and temperature dependence of the photoluminescence spectrum.

The first priority is analysis of the $\text{Cs}_2\text{AgInCl}_6$ study. The self-energy based distortion has provided the starting point for relaxation with excited-state forces. Next year will focus on comparing the relaxed structure with the harmonic self-energy prediction and quantifying the role of nonlinear lattice relaxation.

The second priority is to complete manuscript quality analysis of the LiF and SiO_2 benchmarks. These systems provide a controlled test of the theory against prior calculations and available experimental data. The remaining work is mainly convergence analysis with respect to momentum grid, exciton basis, phonon modes, and the tadpole and Fan Migdal terms.

The third priority is selective extension. Related materials or defect and doping configurations will be pursued only if the current analysis shows that they are needed to strengthen the scientific conclusions. Resource needs may increase for those cases because the effective problem size grows quickly and the *GW* plus BSE stages become more expensive.

The planned work is directly tied to the project milestones:

- Finalize convergence analysis for the finite momentum exciton and self-energy calculations on the benchmark and target systems.
- Complete any targeted refinement of the $\text{Cs}_2\text{AgInCl}_6$ excited-state forces result.

- Compute phonon resolved self-energy contributions, Stokes shifts, and temperature dependent emission linewidths.
- Compare harmonic self-energy predictions with nonlinear results from excited-state forces to define the accuracy and limits of the method.
- Prepare manuscripts on temperature dependent STE emission and the self-energy guided $\text{Cs}_2\text{AgInCl}_6$ study.

No major change in scope is planned. The approach is now more integrated than originally envisioned because the self-energy method is used both for dynamical STE properties and for improving convergence of the excited-state forces method.

2.2 Development Work

Substantial development was completed during the current allocation period. The main effort was the theory and implementation of the STE self-energy method, including the self consistent tadpole term, the Fan Migdal term, finite momentum exciton inputs, and post processing for STE observables. Development also included tests on LiF, debugging, extension to denser grids, and the Frontier build of PETSc, SLEPc, petsc4py, and slepc4py with ROCm and HIP support.

About 60 to 70 percent of the current effort went to development, testing, and validation because the work required new theory, new code, and a reliable bridge to excited-state forces. That development phase is largely complete. Next year’s allocation should therefore be used mostly for analysis, final convergence checks, and manuscript preparation.

Remaining development will focus on robustness and reproducibility. Planned tasks include automated convergence checks, a cleaner interface to BerkeleyGW outputs, lower memory use for dense grids, and reproducible scripts for spectra, linewidths, and mode resolved analysis. Development computing is expected to be about 25 percent of next year’s use, with the rest directed to analysis and targeted production reruns.

2.3 New Code Applications

No entirely new production application is planned for next year. The work will continue to use Quantum ESPRESSO, EPW, BerkeleyGW, and the project specific STE self-energy code. The code stack is already in place; next year’s use will focus on final analysis, selected reruns, and optional extensions rather than a new application path.

The current year’s tests provide the relevant performance evidence. The code has run on Frontier for $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grids using distributed PETSc and SLEPc objects and parallelism over q points. This maps directly to the proposed calculations, where cost grows with the number of momentum points, exciton basis size, and self-energy matrix operations. BerkeleyGW also remains suitable because finite momentum BSE calculations can be batched over independent momentum points.

If defect or doping calculations are added, the main added requirements will be memory and storage. Larger structural cells increase the electronic and excitonic basis sizes, and the GW plus BSE data products grow quickly. The plan is to keep the current application stack and use careful batching and data reduction, rather than introduce a new untested code path.

3 Publications Resulting from INCITE Awards

There are no submitted or published papers yet from this INCITE work that include the required INCITE or Leadership Computing Facility acknowledgement. Two manuscripts are in preparation from the calculations completed so far:

- A paper on the first principles exciton self-energy framework for STEs, including dynamical effects and temperature dependent photoluminescence linewidths.
- A paper on $\text{Cs}_2\text{AgInCl}_6$, using self-energy guided excited-state forces to obtain the STE distortion, Stokes shift, and emission properties of the double perovskite.

These papers will include the required INCITE and facility acknowledgements when submitted.